Experiment 4a:

Code:

```
 GNU nano 9.0                                                exp4a.zsh
#!/bin/zsh

echo "Currently working users are"
echo ""
who
echo ""

read -p "Enter username: " user

login_time=$(who | awk -v u="$user" '$1 == u {print $3, $4; exit}')
if [ -z "$login_time" ]
then
echo "User is not currently logged in"
else
echo "login time: $login_time"

login_seconds=$(date -d "$login_time" +%s)
current_seconds=$(date +%s)

total_seconds=$((current_seconds - login_seconds))
hours=$((total_seconds / 3600))
minutes=$(((total_seconds % 3600) / 60))
echo "Total  working time: $hours hours $minutes minutes"
fi
```

Output:

```
┌──(kali㉿sourav)-[~]
└─$ nano exp4a.zsh

┌──(kali㉿sourav)-[~]
└─$ bash exp4a.zsh
Currently working users are

kali     seat0        2026-09-16 08:55 (:0)

Enter username: kali
login time: 2026-09-16 08:55
Total  working time: 0 hours 30 minutes

┌──(kali㉿sourav)-[~]
└─$
```

Experiment 4b:

Code:

```
#!/bin/zsh

if [ "$#" -ne 3 ]
then
echo "Usage: $0 filename start_line end_line"
fi

file="$1"
start="$2"
end="$3"

sed -n "${start},${end}p" "$file"
```

Output:

```
┌──(kali㉿sourav)-[~]
└─$ nano exp4b.zsh

┌──(kali㉿sourav)-[~]
└─$ bash exp4b.zsh exp4btest.txt 3 6
3
4
5
6

┌──(kali㉿sourav)-[~]
└─$
```

Experiment 4c:

Code:

```
#!/bin/zsh

if [ "$#" -lt 2 ]
then
echo "Need atleast two arguments"
fi

word="$1"
shift

for file in "$@"
do
if [ -f "$file" ]
then
sed -i "/$word/d" "$file"
echo "Deleted lines containing '$word' from $file"
else
echo "Error: $file does not exist or is not a regular file."
fi
done
```

Output:

```
┌──(kali㉿sourav)-[~]
└─$ nano exp4c.zsh

┌──(kali㉿sourav)-[~]
└─$ bash exp4c.zsh apple exp4ctest.txt
Deleted lines containing 'apple' from exp4ctest.txt

┌──(kali㉿sourav)-[~]
└─$ cat exp4ctest.txt
banana
orange
```